CSA15 Cloud Computing and Big Data Analytics

Local IaaS Operations Lab: Snapshot, Clone and Restore Testing

Course Code	CSA15	**Course Title**	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-03	**Submission**	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	**Employee ID**	SSETSCS415
Target Learner	Beginner CSE student	**Estimated Duration**	2 to 3 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation	**Cloud Link**	Builds operational recovery thinking before cloud VM backup and image management
Guided Project Pattern Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.			

Learning Outcomes

- Create a safe baseline snapshot of a configured VM.
- Test snapshot recovery by making a visible temporary change and restoring the previous state.
- Create or plan a full/linked clone according to storage availability and tool support.
- Compare snapshot, clone and VM image/export in operational terms.
- Prepare a complete VM recovery evidence portfolio.

Before You Start

- Completed VM from GP-VM-01 and configuration/storage evidence from GP-VM-02.
- At least 5 GB free storage for clone testing where possible; otherwise prepare a justified clone plan.
- The VM must not contain passwords, keys or personal sensitive data in screenshots.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A snapshot records a point-in-time VM state and is useful before risky changes.
- A clone creates another VM copy for parallel testing or repeated experiments.
- A full clone is independent and consumes more storage. A linked clone depends on the base VM and usually consumes less storage. - A restore test is important because a snapshot is useful only if it can actually return the VM to an earlier state.

Lab Exercise Coverage

Experiment 10	Create a snapshot and test it by loading/restoring the previous version.
Experiment 11	Create a clone of a VM and test it by opening the cloned VM.

Hypervisor Choice

Oracle VirtualBox	Use Snapshots from the VM tools/menu and Clone from Machine options.	Snapshot list, restore test, clone window and cloned VM proof.
VMware Workstation	Use Snapshot Manager / Take Snapshot and VM -> Manage -> Clone where available.	Snapshot manager, revert proof, clone wizard and cloned VM proof.

Safety Rule

Do not delete the original VM while testing snapshots or clones. If storage is low, record a clone plan with menu evidence instead of creating a large duplicate.

Architecture View

Base VM	Baseline Snapshot	Temporary Change	Restore Test	Clone/Plan	Recovery Evidence

The recovery workflow begins with a stable VM, captures a baseline, performs a controlled change, restores the previous version and then prepares a duplicate environment or clone plan.

Guided Build

Stage 1 - Prepare the Baseline VM

Learn: Snapshot and clone testing must begin from a known working VM state.

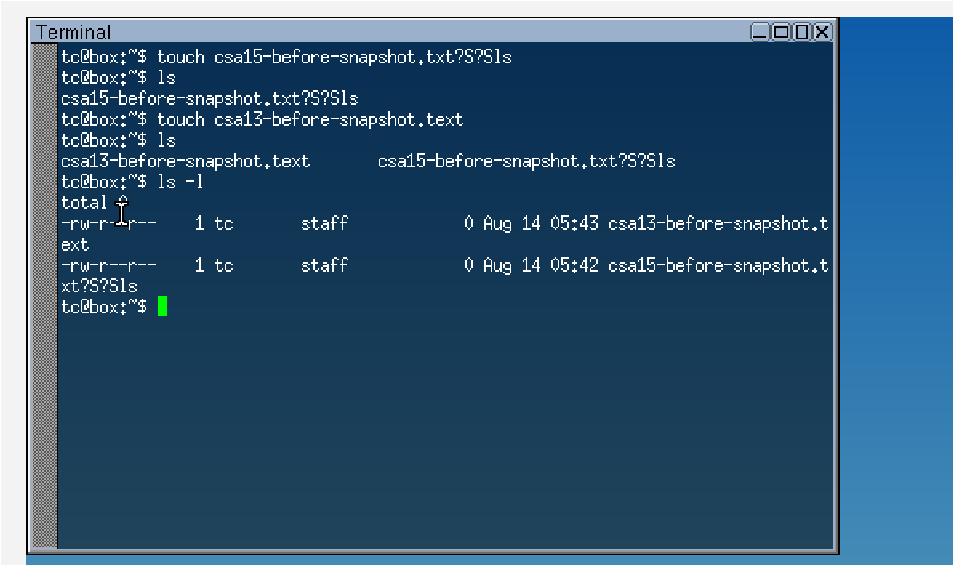
- Open the VM in VirtualBox or VMware.
- Boot the guest OS and confirm that it reaches desktop, shell or login prompt.
- Create a visible marker file inside the guest OS if possible, such as csa15-before-snapshot.txt.
- Shut down or pause the VM according to the hypervisor recommendation.

Checkpoint

A stable baseline VM is ready and the before-snapshot state is visible.

Record 1 - Baseline state evidence

Observation: The Tiny Core Linux virtual machine was booted successfully and a baseline marker file named csa15-before-snapshot.txt was created. The ls -l command confirmed that the file exists in the home directory. This establishes the initial stable state of the VM before taking the snapshot.



Stage 2 - Take a Snapshot

Learn: A snapshot preserves the current VM state before a risky action.

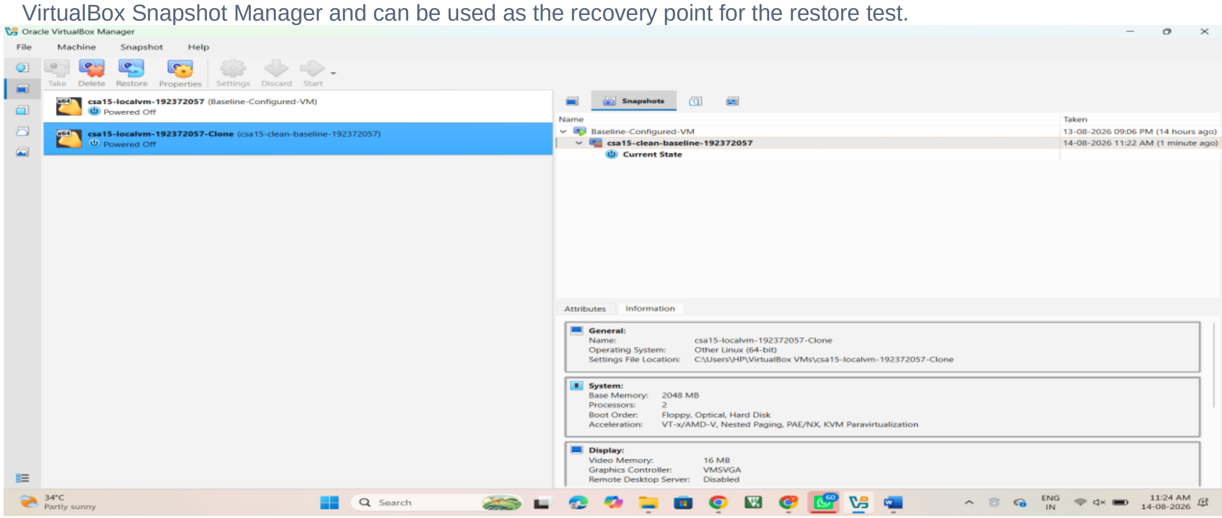
- In VirtualBox, select the VM and open Snapshots, then choose Take.
- In VMware, select VM -> Snapshot -> Take Snapshot, or open Snapshot Manager.
- Name the snapshot csa15-clean-baseline-register-number.
- Write a short description: stable VM before restore test.
- Capture the snapshot list or manager screen.

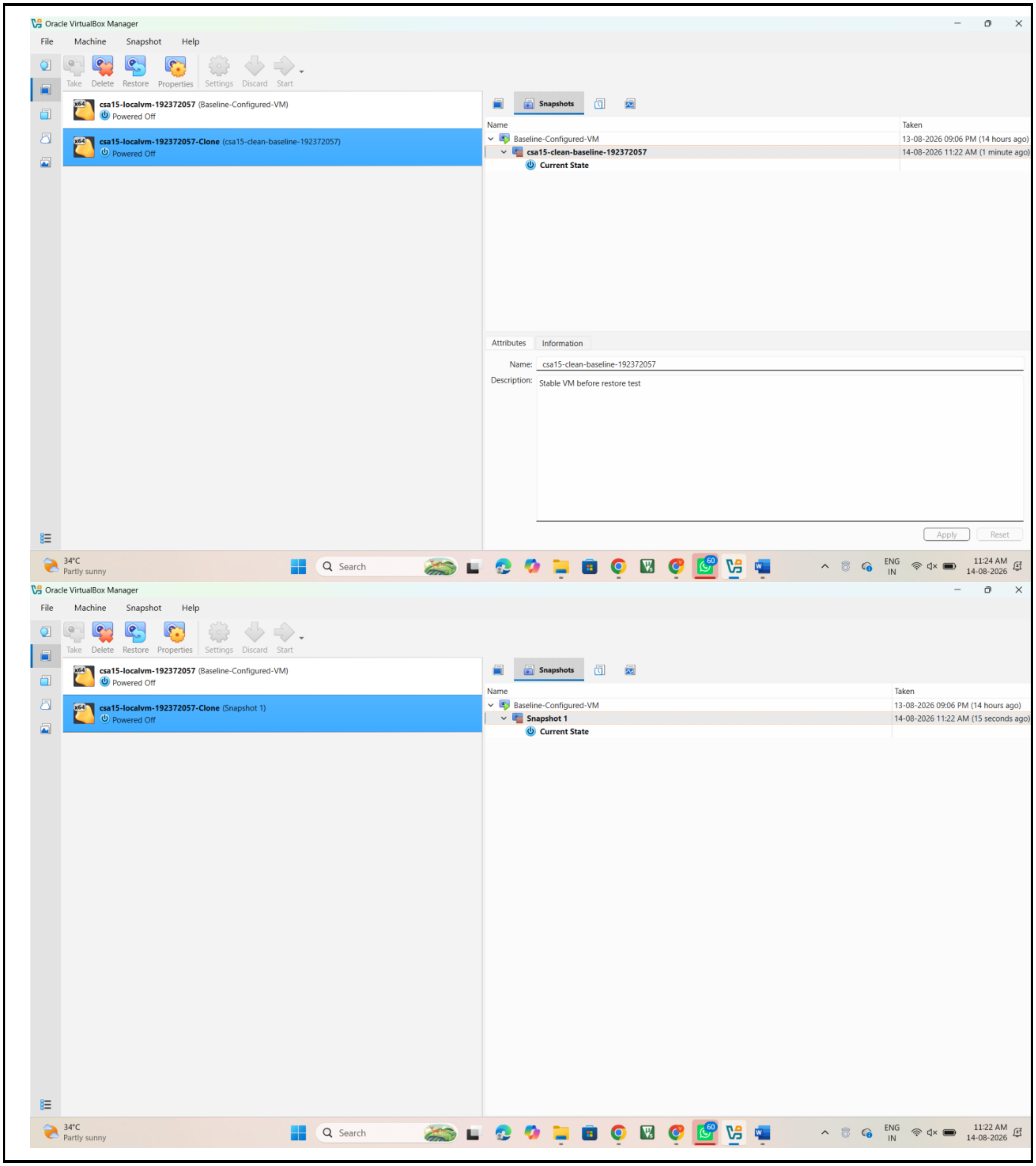
Checkpoint

The named baseline snapshot is visible in the hypervisor.

Record 2 - Snapshot creation evidence

Observation: A baseline snapshot named csa15-clean-baseline-192372057 was created successfully while the VM was in its stable initial state. The description was set to “Stable VM before restore test”. The snapshot is visible in the VirtualBox Snapshot Manager and can be used as the recovery point for the restore test.





Stage 3 - Make a Controlled Change and Restore

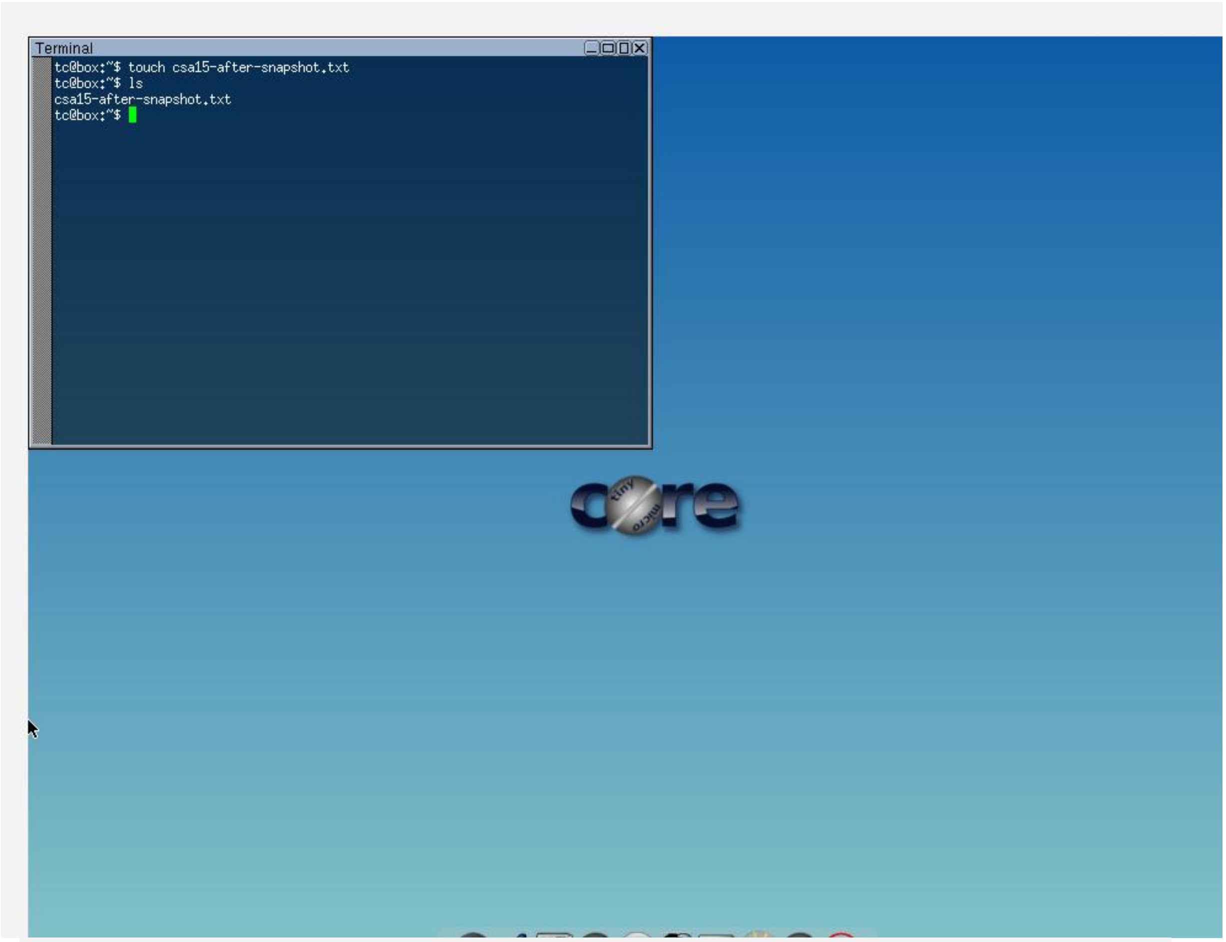
Learn: A restore test proves that the snapshot is usable.

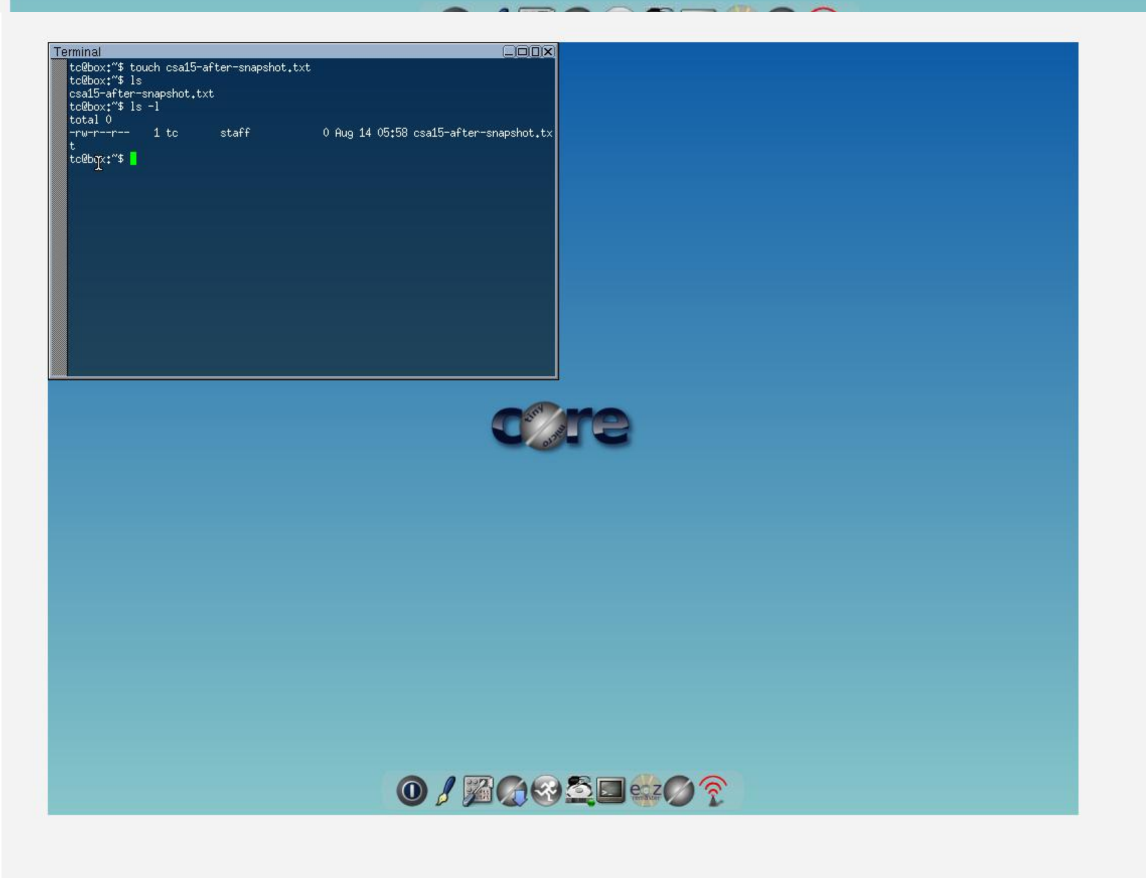
- Boot the VM after snapshot creation.
- Create a temporary visible change such as csa15-after-snapshot.txt, a desktop folder or a small configuration note. - Capture evidence of the temporary change.
- Shut down or pause the VM and restore/revert to the baseline snapshot.
- Boot again and verify whether the temporary change disappeared or the previous state returned.

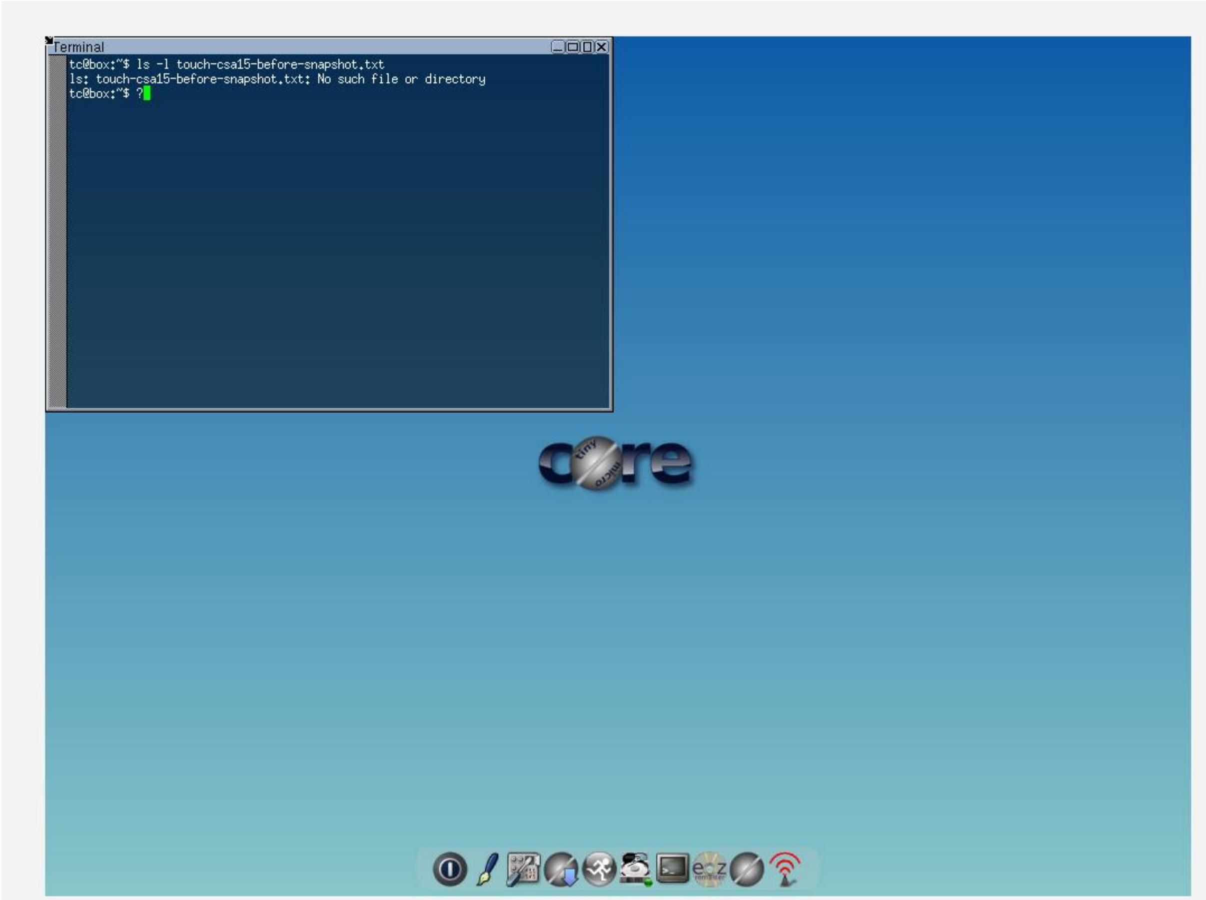
Checkpoint

The restore result is documented with before-change, after-change and after-restore evidence.

Record 3 - Snapshot restore test evidence







Stage 4 - Create or Plan a Clone

Learn: A clone creates a second VM environment that can be used without disturbing the base VM. - In VirtualBox, choose Clone from Machine tools/menu and name it CSA15-Clone-RegisterNumber.

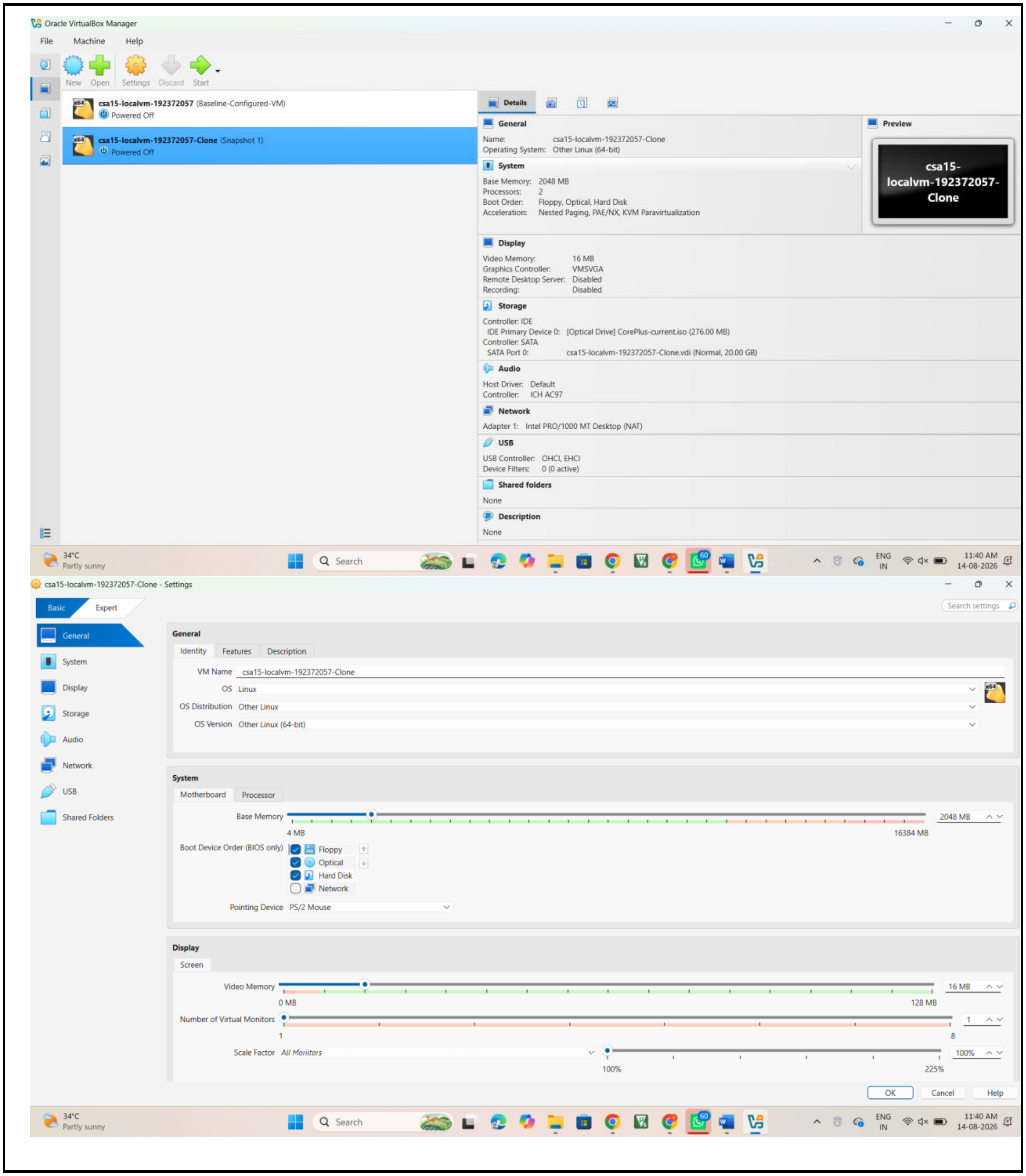
- In VMware, choose VM -> Manage -> Clone if the installed edition supports it.
- Choose full clone if sufficient storage is available. Choose linked clone only when the tool supports it and the base VM will remain available.
- If the tool edition or storage does not allow cloning, record the exact limitation and write a clone plan.
- Open the cloned VM or show the completed clone/plan evidence.

Checkpoint

The clone is created and visible, or a technically justified clone plan is completed.

Record 4 - Clone evidence or clone plan

A clone of the baseline virtual machine is already available in Oracle VirtualBox. The existing clone provides a separate VM environment for testing without modifying the original VM. A full clone is useful when an independent copy is required for experimentation and recovery.



Stage 5 - Build the Recovery Portfolio

Learn: Professional VM operations require evidence, decision reasoning and cleanup status. - Prepare a comparison table: snapshot, full clone, linked clone, exported image and rebuild from ISO.

- Record storage impact: original VM size, snapshot impact and clone/export estimate where visible.
- Write when you would use snapshot, clone or rebuild for your capstone project.
- Remove only temporary cloned/exported files created for this lab if storage is low and after evidence is captured.

Checkpoint
The portfolio proves snapshot creation, restore testing, clone/plan and capstone recovery reasoning.

Record 5 - Final recovery portfolio

The virtual machine recovery process was tested using a snapshot and an existing clone. The snapshot provides a point-in-time recovery state for the VM, while the clone provides a separate VM environment for testing without disturbing the original VM. The clone was verified in Oracle VirtualBox with its VM configuration and storage details.

Recovery Method	Purpose	Storage	Use
Snapshot	Saves a point-in-time VM state	Lower than a full clone	Quick recovery before risky changes
Full Clone	Creates an independent VM copy	Higher	Separate testing environment
Linked Clone	Depends on the base VM	Lower	Testing when storage is limited
Exported Image	Portable VM copy	Depends on image size	Backup/transfer
Rebuild from ISO	Creates VM again from installation media	Requires installation/configuration	When recovery copy is unavailable

The original VM has a 20 GB virtual disk. An additional 2 GB virtual disk was created during the earlier virtual-disk exercise. The clone provides a separate VM environment and therefore requires additional storage compared with using only a snapshot.
A snapshot would be used before making risky configuration or software changes to the capstone environment. A clone would be used when a separate testing environment is required. An exported VM image would be useful when a portable recovery copy is required. Rebuilding from ISO would be the final option when an existing recovery copy is unavailable.

Final Evidence Checklist

□	**Baseline prepared**	Stable VM and before-snapshot marker evidence.
□	**Snapshot created**	Snapshot manager/list screenshot with correctname.
□	**Restore tested**	Before-change, after-change and after-restore evidence.
□	**Clone completed/planned**	Clone proof or justified clone limitation/plan.
□	**Recovery reasoning written**	Snapshot vs clone vs image/export comparison and capstone link.

Explain and Extend

Explain why a snapshot is not the same as a full backup.

A snapshot records the state of a virtual machine at a particular point in time and is mainly useful for quick recovery after changes. It should not be treated as a complete independent backup because it depends on the VM's storage and snapshot chain.

Explain when a full clone is better than a linked clone.

A full clone is better when an independent VM is required. It does not depend on the original VM and can be used separately for testing or experimentation. A linked clone is more suitable when storage space is limited and the base VM will remain available.

Explain how restore testing reduces risk during capstone development.

Restore testing confirms that the recovery point actually works. It allows changes to be tested safely because the VM can be returned to a known working state if a configuration or software change causes a problem.

Extend this setup to your capstone: what change would you test only after taking a snapshot?

I would take a snapshot before making major operating-system configuration changes, installing new software packages, changing network settings, or modifying important project dependencies. If the change causes a problem, the VM can be restored to the previous working state.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM03-Snapshot-Clone-Restore-Test.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status. - Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: K.R.VISHNU CHAITHANYA Register Number: 192372057 Date: 14-08-2026